Integrate out the cavity photon, bin the phonons

Lunkin, Holubeu, Golež and Lenarčič do not solve the steep-photon problem, because their model never runs into it. Their closed form (Eq. 16) exists only because each bath rate splits into two single-leg Fermi factors, with no energy or momentum conservation linking the legs. The authors say themselves that Fabry–Pérot cavities "will necessitate tailored modifications of the system–bath coupling" (arXiv:2609.10062). The clean fix comes from the cavity-superconductor and exciton/Floquet literature, which always integrates out the photon's in-plane momentum analytically before the electron grid is involved. With $c \approx 3950$ in lattice units, every resonant photon has $q \lesssim 6 \times 10^{-4}\, a^{-1}$, so all photon transitions are vertical. Doing the q integral with the planar-cavity density of states turns the narrow Lorentzian into a smoothed step above ω_0 (TE1+TM1) and above $2\omega_0$ (TM2). The photon bath then becomes a pair-creation/annihilation rate that is local in k, with closed-form arctan and complex-log kernels. Your current Cartesian code instead samples one Lorentzian in the $q = 0$ cell with weight $1/N^2$. At $N = 64$–256 that overweights the photon bath by 10^2–10^5 near resonance and gives the wrong spectral shape; reaching convergence would take $N \approx 5 \times 10^5$. The cold acoustic phonons ($\omega_D \approx 0.16$–$0.22\, t_a$, below 2Δ) are quasi-elastic, and with sharp lines they cannot recombine pairs. Fokker–Planck holds only for $T_{\mathrm{cold}} \gg \omega_D$, so the default should be an exact kernel that is banded in energy. There is no rank-one analogue of Eq. 16. The problem does admit three useful forms: an exact per-k quadratic update that contains Eq. 16 as a special case, a fast-phonon quasi-Fermi state (the Owen–Scalapino μ^* state) fixed by two global constants, and a relaxation-time closure. Physically, hot vertical photons only break pairs. They cannot push quasiparticles up in energy, which is what stabilises order in the paper. Order is therefore enhanced by photons colder than the phonons, not hotter, which agrees with Curtis et al. The first code change should be to take the photon bath out of the q-indexed line table and put it back as a local term.

Throughout, paragraphs headed **Sources** report only what the cited papers state. Paragraphs headed **Derivation** or **Inference** are reasoning by the research team or by this synthesis, and are unpublished. Several of them were checked numerically, as noted where they appear.

Lunkin's Eq. 16 needs a white, separable, momentum-free bath

Sources.

Model and rates. The paper takes a mean-field BCS Hamiltonian with a Lindblad dissipator $\gamma_0[\hat{D}^{(T_1)} + \hat{D}^{(T_2)}]$. Its jump operators $L^{(s_1,s_2)}_{k_1,k_2} = \sqrt{\gamma^{(s_1,s_2)}_{k_1,k_2}}\, b^{(s_1)}_{k_1} b^{(s_2)}_{k_2}$ cover pair creation, pair annihilation and scattering. The only physical requirement placed on the baths is detailed balance. Following Zanoci–Yoo–Swingle, the authors then choose the factorised rates $\gamma^{(s_1,s_2)}_{k_2,k_1}(T) = \alpha_{s_1 s_2} f_{s_1}(\epsilon_{k_1}/T) f_{s_2}(\epsilon_{k_2}/T)$ (Eq. 13) (arXiv:2609.10062; Zanoci et al.).

Closed form. With these rates the stationary occupation is $n_k = \sum_j f_+(\epsilon_k/T_j) A(T_j) \big/ \sum_j [f_+(\epsilon_k/T_j) A(T_j) + f_-(\epsilon_k/T_j) B(T_j)]$, where A and B are k-independent sums over the partner state (Eqs. 14–16). For $\alpha = 1$ this reduces to $n_k = [f_+(\epsilon_k/T_1) + F f_+(\epsilon_k/T_2)]/(1+F)$, with one self-consistent weight F (Eqs. 17–18).

Coupling-independence. **The coupling γ_0 does not enter the steady state**, so the deviation from thermal occupation is an O(1) "greenhouse effect". The paper contrasts this with the Eliashberg effect and its cavity extension, where the response scales with the inelastic rate (arXiv:2609.10062).

Numerics. The s-wave case is a fixed-point iteration: update n with Eq. 16, then update the gap from $\ln(\Delta_0/\Delta_{\mathrm{new}}) = 2\int d\xi\, n/\epsilon$, on a ξ grid with cutoff $\Lambda = 10\Delta_0$ and $L = 1000$ points, until the gap changes by less than $10^{-4}\Delta_0$. The d-wave case is time propagation on a 1600×1600 grid.

Scope. The authors say their baths induce "transitions across a large fraction of the Brillouin zone", a regime they associate with surface-plasmon or hyperbolic polaritons. They say Fabry–Pérot and split-ring cavities need tailored couplings, and they list excitonic condensates as an open target. They also speculate that their second-order transition may be a consequence of Markovian baths (arXiv:2609.10062).

Framework papers. In the earlier work from the same group, the GGE multipliers are fixed by conditions from which the coupling strength drops out, yet the multipliers still shift at O(1) (Lenarčič–Lange–Rosch, arXiv:1706.05700). The time-dependent GGE needs a unique, stable fixed point of the rate equations and carries $O(\epsilon^2)$ corrections (Lange–Lenarčič–Rosch).

Derivation.

Why the closed form exists. Insert Eq. 13 into the Pauli equation. The gain into k becomes $f_+(\epsilon_k/T_j)(1-n_k)$ times a bracket summed over k', and the loss becomes $f_-(\epsilon_k/T_j) n_k$ times another such bracket. Neither bracket depends on k. So for each process and each bath the kernel is **rank one in (k, k')**, and stationarity is linear in n_k with coefficients that depend on ϵ_k alone. This needs three things together:

1. the rates depend on momentum only through ϵ;
2. the rate factorises over the two legs;
3. no energy-conserving $\delta(E_k \pm E_{k'} - \Omega)$ links the legs, since without that condition the factorisation in (2) is

impossible.

Momentum independence alone gives only an energy-only equation, not a closed form. More generally, any rank-r kernel $\sum_m a_m(\epsilon_k)b_m(\epsilon_{k'})$ gives a closed form with $2r$ global numbers per bath.

Golden-rule reading. The research team checked these identities numerically: $f(x)f(y) = N(x+y)[1-f(x)-f(y)]$ and $f(x)[1-f(y)] = [f(x)-f(y)][1+N(y-x)]$. So Eq. 13 is a bosonic golden rule whose effective spectral density is $J_{\text{eff}} = 1-f(x)-f(y)$ for pair processes and $f(x)-f(y)$ for scattering. This J_{eff} depends on the bath temperature and selects no energy. A cavity photon or an acoustic phonon is different: it has a temperature-independent $J(\Omega)$ multiplying an energy delta (or Lorentzian), together with momentum conservation, and neither of these factorises. **The closed form is a property of an engineered all-to-all, white bath, not of the GGE method.**

Coupling ratio. If the two baths couple with different strengths $g_1 \neq g_2$, then $F \to (g_2/g_1)A(T_2)/A(T_1)$. Only the overall scale drops out; the coupling ratio stays a physical parameter. This matters for your model, where the photon and phonon amplitudes are independent.

Explicit F. For $\alpha = 1$, write $a = 2N^{11}_{+-}$, $d = 2N^{22}_{+-}$ and $b = N^{12}_{+-} + N^{12}_{-+}$. Eq. 18 then becomes $bF^2 + (a-d)F - b = 0$, so $F = [(d-a) + \sqrt{(d-a)^2 + 4b^2}]/2b$.

What carries over to your model:

- the GGE in μ_k, which you already use;
- independence from the overall coupling scale;
- the iteration pattern: a closed-form local update, a few global functionals, then a gap update;
- the ξ-variable gap integral, which removes the BCS edge singularity;
- the vanishing of the first term of Eq. 8, which the research team argues carries over to the EI. Its GGE expectation contains only $\alpha^\dagger_k\beta_k$-type terms, which are zero.

Every comparable kinetic theory integrates the photon momentum away

No paper in the surveyed literature resolves a narrow photon line on an electron momentum grid. Each one does one of two things:

- integrates the in-plane photon momentum analytically into a spectral density (a density of states), or
- sets the photon momentum to zero and assigns each k an analytic rate.

When the rates then depend on k only through energy, the electronic problem becomes energy-only and is solved on a 1D grid of about 10^3 points.

Line of work	Photon treatment	Cold bath / phonons	Grid and solver
Lunkin et al. 2026 (arXiv)	No spectral or momentum structure (Eq. 13)	Same factorised form; optional RTA γ_1	1D ξ grid, $L = 1000$, fixed point; d-wave 1600^2, time propagation
Curtis et al. 2019 (arXiv)	q integrated analytically, $q\,dq = \omega' d\omega'/c^2$, giving a multimode kernel; momentum non-conservation from disorder	RTA τ_{in} toward $n_F(E/T_{\text{qp}})$	Energy-only, linear response $\delta n = \tau_{\text{in}} I_{\text{cav}}$
Piazza group 2025 (PRR 7, 013073; arXiv:2509.07865)	Photon momentum set to zero; delta or Lorentzian line	Wide-band "cryostat" damping at T_{cry}	Energy-only, iterative
Chang–Scalapino / Goldie–Withington / Guruswamy (arXiv:1208.0685; arXiv:1401.1937)	Monochromatic $q = 0$ line; strength B fixed by absorbed power	Full $\alpha^2F = b\Omega^2$ kernel plus phonon equation	Uniform 1D E grid, 1000–2500 bins; damped Newton, ~10 iterations
Floquet two-bath (Seetharam 2015; Esin 2018)	Vertical radiative recombination, $\delta_{kk'}$, constant photon DOS	3D phonons, q_z integrated analytically	50×50 k grid, Newton
Exciton/polariton (Tassone 1997; Ivanov 1999; Porras 2002)	Radiative rate analytic per state (light-cone formulas)	Deformation potential, delta removed by a Jacobian	Energy or $\lvert k\rvert$ grids, uniform/log/adaptive

Sources.

Curtis et al. (cavity Eliashberg). This is the closest analogue to your setup. A 2D film sits at the antinode $z = L/2$ of a planar cavity with $\omega_q = \sqrt{\omega_0^2 + c^2 q^2}$, coupled by $j \cdot A$. At the sheet, the polarization vectors sum and angle-average to $(2/L)(1 + \omega_0^2/\omega_q^2)$. The appendix kernel is $J_{\mathrm{MM}}(\omega) = (2\kappa/L)\int_{\omega_0}^{\infty} d\omega' [\frac{1}{(\omega-\omega')^2+\kappa^2} - \frac{1}{(\omega+\omega')^2+\kappa^2}](1 + \omega_0^2/\omega'^2)$, which is "roughly constant for $\Omega > \omega_0$". A single-mode cavity, by contrast, gives a Lorentzian at ω_0 (arXiv:1805.01482v3). Curtis et al. reach an energy-only equation by impurity-assisted momentum non-conservation in the strong-disorder limit, not by assuming $q = 0$ (APS accepted manuscript).

Piazza group. They set the photon momentum to zero "because cavity photons are very light compared with electrons" (PRR 7, 013073). They keep only $q = 0$ forward scattering, with a Lorentzian photon spectral function and impurity-induced momentum non-conservation. They find a bath-induced cubic Ginzburg–Landau term that makes the transitions first order (arXiv:2509.07865).

Planar microcavity momentum spread. Chakraborty and Piazza note that transverse modes of a planar microcavity "are spread around $q = 0$ with width ω_c/c". Their single-mode results carry over if this momentum broadening is smaller than the electron lifetime (ar5iv 2008.06513).

Eliashberg / Chang–Scalapino. The classic superconducting kinetics never resolve photon momentum either. The microwave enters as a pair of shift terms $f(E) \leftrightarrow f(E \pm \hbar\omega)$ with the factor $1 + \Delta^2/EE'$. A third, pair-breaking term appears for $\hbar\omega \geq 2\Delta$ (arXiv:1208.0685; arXiv:1401.1937).

Goldie–Withington. They put photon energies on bin multiples "to place photon peaks in well-defined bins", and solve with damped Newton and an analytic Jacobian in about 10 iterations (arXiv:1208.0685).

Floquet two-bath kinetics. Seetharam et al. make radiative recombination strictly vertical, $g^{\mathrm{rec}}\delta_{q_x,0}\delta_{k,k'}$, with a constant photon DOS (arXiv:1502.02664). Esin et al. lump all photon final states within $|q| \lesssim \Omega/v_\ell$ onto k, with multiplicity $L_x L_y \Omega^2/(4\pi v_\ell^2)$, because the rates are smooth on that scale (arXiv:1710.09404). That is exactly the "q = 0 cell integral" argument used below.

Excitons and polaritons. Exciton codes assign radiative rates analytically per state, for example $\Gamma_T = \Gamma_0 k_0/\sqrt{k_0^2 - p^2}$ inside the light cone (Ivanov et al.). Polariton codes use $|C_k|^2/\tau_c$ (Porras et al.).

Brute-force alternatives. A flat-band microwave study broadens the energy delta into a 0.1 meV Lorentzian on a 50×50 grid (arXiv:2411.03437). Ab initio Boltzmann codes warn that "convergence with respect to both the q-point grid and broadening needs to be checked carefully" (Perturbo, arXiv:2002.02045). Snoke notes that Lorentzian broadening of energy conservation "has not been well theoretically justified" and can destabilise the total energy (arXiv:1011.3849).

Cavity-coupled EI papers. These are all equilibrium or coherent calculations: a single $q = 0$ mode in the superradiant EI (Mazza–Georges), an RPA photon propagator (Davari–Kargarian), and a driven bilayer with leads (Sun et al.). **None has a cavity-bath kinetic equation.**

Inference. The cleanest template is Curtis's appendix kernel combined with Seetharam's vertical $\delta_{kk'}$. The electronic factors are evaluated at $k' = k$, and the photon q integral is done in closed form. Curtis needed disorder only because in a clean single-band BCS superconductor the $q = 0$ pair-breaking current element vanishes. In your two-orbital EI it does not, as the next section shows, so the clean vertical limit is non-trivial.

The q = 0 cell already holds the entire photon continuum

Derivation: kinematics and selection rules (research team; numbers checked here). In lattice units, $c = \hbar c/(t_a a) = 197.3\,\mathrm{eV\,nm}/(0.1\,\mathrm{eV} \times 0.5\,\mathrm{nm}) = 3946$.

- For $\omega_0 = 1.6$, the resonant in-plane momenta are $q_{\mathrm{res}} = \sqrt{\Omega^2 - \omega_0^2}/c = 1.0 \times 10^{-4}, 3.0 \times 10^{-4}$ and 6.4×10^{-4} at $\Omega = 1.65$, 2 and 3. The grid spacing at $N = 64$ is $2\pi/N \approx 0.098$.
- The recoil mismatch $v \cdot q \lesssim 10^{-3}$ is far below γ and T, so the electronic factors can be set to $k' = k$.
- **Interband:** only $\beta_k \to \alpha_k$ absorption and $\alpha_k \to \beta_k$ emission at $\Omega = 2E_k$ survive, and the band-centre term $\bar{E}(s)$ cancels exactly.
- **Intraband:** a transition would need $E(k+q) - E(k) = \omega_q \geq \omega_0$ at $|q| \sim 10^{-4}$. This is impossible because $v_{\max} \approx 2.8 \ll c$. On the grid, the nearest distinct final state has $\omega_q \geq c\,2\pi/N \approx 390$, so the intraband tails are suppressed by roughly 10^{-9}.
- **Conservation law:** vertical transitions conserve $n_\alpha(k) + n_\beta(k)$ separately at every k. A photon-only bath is therefore non-ergodic, and its Jacobian is singular. Phonons must fix these N_k quantities.

Derivation: the continuum integral. With $(1/N_k)\sum_q \to \int d^2q/(2\pi)^2 = (2\pi c^2)^{-1}\int_{\omega_{\min}}^{\infty} \omega\, d\omega \langle\cdot\rangle_\phi$, the $1/\omega_q$ in the mode normalisation cancels the 2D massive-photon DOS. Write $\rho_\omega(\Omega) = \pi^{-1}[\gamma/((\Omega-\omega)^2+\gamma^2) - \gamma/((\Omega+\omega)^2+\gamma^2)]$. Your line factor is exactly $2\pi N(\Omega)\rho_{\omega_q}(\Omega)$. The polarization weights at $z = L/2$ are:

Modes	Field component at $z = L/2$	Weight $\langle P \rangle_\phi$	Threshold
TE1 + TM1	in-plane	$\lvert C_1\rvert^2 \frac{\lvert X_k\rvert^2}{2}(1+\omega_0^2/\omega^2)$	ω_0
TM2	E_z (couples to the z-dipole)	$\lvert C_2\rvert^2(1-4\omega_0^2/\omega^2)$	$2\omega_0$

Here $|X_k|^2 = \sin^2 k_x + \sin^2 k_y = 2 - (s^2+t^2)/2$.

Derivation: closed-form kernels. The ω integrals have closed forms:

$$K_0(\Omega; a) = \int_a^\infty \rho_\omega \, d\omega = \tfrac{1}{\pi}\Big[\arctan\tfrac{\Omega-a}{\gamma} + \arctan\tfrac{\Omega+a}{\gamma}\Big],$$

$$K_2(\Omega; a) = \int_a^\infty \frac{\rho_\omega}{\omega^2} d\omega = \tfrac{1}{\pi}\mathrm{Im}[F(z) - F(-\bar{z})], \quad z = \Omega + i\gamma, \quad F(z) = -\frac{\ln(1-z/a)}{z^2} - \frac{1}{az}.$$

These were re-verified here against adaptive quadrature to seven digits, including below, at and above threshold. In the sharp limit $K_0 \to \theta(\Omega - a)$ and $K_2 \to \theta(\Omega - a)/\Omega^2$.

The in-plane combination $K_0 + \omega_0^2 K_2$ is precisely Curtis's multimode kernel J_{MM} (arXiv:1805.01482v3), evaluated exactly rather than asymptotically.

Derivation: the local rate. The photon bath reduces to

$$\Gamma^{\mathrm{abs}}_{\beta\to\alpha}(k) = \frac{\mathrm{amp}}{c^2} N_h(2E_k)\mathcal{R}(k), \qquad \Gamma^{\mathrm{em}}_{\alpha\to\beta}(k) = \frac{\mathrm{amp}}{c^2}[1 + N_h(2E_k)]\mathcal{R}(k),$$

$$\mathcal{R}(k) = |C_1^{\alpha\beta}|^2 \tfrac{|X_k|^2}{2}\big[K_0(2E_k;\omega_0) + \omega_0^2 K_2(2E_k;\omega_0)\big] + |C_2^{\alpha\beta}|^2\big[K_0(2E_k;2\omega_0) - 4\omega_0^2 K_2(2E_k;2\omega_0)\big].$$

It enters the kinetic equation as the local current $J_{\mathrm{ph}}(k) = \Gamma^{\mathrm{em}} n_\alpha(1-n_\beta) - \Gamma^{\mathrm{abs}} n_\beta(1-n_\alpha)$.

- **Detailed balance at T_h is exact**, since $\Gamma^{\mathrm{abs}}/\Gamma^{\mathrm{em}} = e^{-2E_k/T_h}$ identically.
- The coherence factors follow from the conventions in your project document `EI_bins_design.md`. At $k = k'$ they give $C_1^{\alpha\beta} = 2(t_a - t_b)u_k v_k = (t_a - t_b)\Delta/E_k$, which is non-zero because the two orbitals have different velocities.
- $C_2^{\alpha\beta} = d\,(e_a - e_b)$ is a function of s alone. **The TM2 term is therefore exactly isotropic within s-shells, while the in-plane term carries the C_{4v}-invariant anisotropy of $|X_k|^2$.**

Inference: what the physics looks like. The 2D continuum turns the "resonant" cavity into a **broadband step absorber**: every k with $2E_k > \omega_0$ is resonant with some in-plane mode. With $\omega_0 \approx 2\Delta$ the step sits at the gap edge. The pumping profile is then set by $N_h(2E_k)$ and by the factor $(\Delta/E_k)^2$ in $|C_1|^2$, and both favour the gap edge. If ω_0 sits slightly above 2Δ, an unpumped ring remains, filled only by the γ tails. Because Δ moves self-consistently, the threshold can move relative to the gap edge. In the sharp limit TM2 switches on only for $E_k > \omega_0 \approx 2\Delta$, and its weight vanishes at threshold as $1 - \omega_0^2/E_k^2$.

Derivation: grid artifact. On a Cartesian grid, the only photon momentum within q_{res} of zero is the $q = 0$ grid point. The code therefore evaluates one Lorentzian centred at $\Omega = \omega_0$, with weight $1/N_k$ per pair. That is a rate scaling as $1/N^2$, with a Lorentzian shape instead of a step. The table gives the ratio of the grid rate to the continuum rate. It was recomputed here for $\omega_0 = 1.6, \gamma = 0.05$, in-plane modes, and identical amp normalisation.

$\Omega = 2E_k$	$N = 64$	$N = 128$	$N = 256$	$N = 1024$
1.6 (on resonance)	2.0×10^5	5.1×10^4	1.3×10^4	795
2.0	1850	462	116	7.2
3.0	173	43	10.8	0.68
5.0	28	6.9	1.7	0.11

- At practical N the grid overweights the photon bath enormously near resonance. At very large N it underweights the high-energy part, because the true step is flat.
- **The photon/phonon rate ratio, which is what sets the NESS, does not converge until $2\pi/N \lesssim \gamma/c$, i.e. $N \gtrsim 2\pi c/\gamma \approx 5 \times 10^5$.**
- If the code normalises per-pair rates differently, the absolute ratio changes, but the $1/N^2$ drift and the shape error do not.

- This agrees with your own measurement in `EI_grid_design.md`. With a much milder steep dispersion ($c_s = 10$), $nk = 40$ was already off by 9% (median) and 28% (maximum) against $nk = 160$, and a cell-averaged Lorentzian "alone does not cure" it. It also explains why $\gamma = 0$ behaved as "effectively a one-cell line width" there.

Inference: what this means for `_ltab`. On a uniform grid, the exact replacement for the $q = 0$ entry is $N_k \int_{\text{cell}} d^2q/(2\pi)^2(\dots)$, i.e. the **average of the full photon factor over the $q = 0$ cell.** With cells of width ~ 0.1 and resonant rings of radius $\sim 10^{-4}$, that cell average equals the continuum integral.

- If `_ltab` sub-samples the cell, no practical sub-grid can resolve the ring.
- If it averages analytically but evaluates the vertex at $q = 0$, it misses the ω-dependent weights. For example, setting $r_2 = (\omega_0/\omega_q)^2 = 1$ instead of keeping it inside the integral overestimates the in-plane rate at $\Omega = 2\omega_0$ by about 60% (a factor of 2 against 1.25 in the sharp limit).
- The photon "line table" should therefore become three functions of Ω evaluated at $\Omega = 2E_k$: $K_0(\Omega;\omega_0)$, $\omega_0^2 K_2(\Omega;\omega_0)$ and $K_0(\Omega;2\omega_0) - 4\omega_0^2 K_2(\Omega;2\omega_0)$.
- All $q \neq 0$ photon entries can be dropped.

A minimal NumPy version, matching the research team's verified code:

python

```python
import numpy as np

def k0(om, a, gam):        # int_a^inf rho_w(om) dw: smoothed step at om = a
    return (np.arctan((om - a) / gam) + np.arctan((om + a) / gam)) / np.pi

def k2(om, a, gam):        # int_a^inf rho_w(om) / w^2 dw: TM weight
    z = om + 1j * gam
    F = lambda z: -np.log(1 - z / a) / z**2 - 1 / (a * z)
    return np.imag(F(z) - F(-np.conj(z))) / np.pi

def r_loc(e, x2, c1, c2, w0, gam):    # R(k); e = E_k, x2 = |X_k|^2
    om = 2 * e
    inp = 0.5 * x2 * c1**2 * (k0(om, w0, gam) + w0**2 * k2(om, w0, gam))
    tm2 = c2**2 * (k0(om, 2 * w0, gam) - 4 * w0**2 * k2(om, 2 * w0, gam))
    return inp + np.maximum(tm2, 0.0)    # clip round-off far below 2 w0
```

What the code does:

- `k0` and `k2` are the two kernels above.
- `r_loc` returns $\mathcal{R}(k)$. Multiply it by `amp / c**2` times N_h or $1 + N_h$ to get Γ^{abs} or Γ^{em}.
- The TM2 integrand is non-negative, so the clip only removes floating-point cancellation.
- For γ below about 10^{-8}, use the sharp-limit formulas instead.

Cold acoustic phonons are quasi-elastic but rarely Fokker–Planck

Derivation: scales.

- $c_s = 0.05$ corresponds to 3.8 km/s.
- The largest phonon energy is $\omega_D = c_s\pi\sqrt{2} \approx 0.22$ at the zone corner (22 meV, 258 K), and $c_s\pi \approx 0.16$ at $(\pi, 0)$.
- Using the NESS gaps $d \approx 0.64$–0.95 recorded in your project notes, $\omega_D/2\Delta \approx 0.12$–$0.17$. **One-phonon interband processes, and so phonon pair recombination, are kinematically forbidden whenever $\Delta > \omega_D/2 \approx 0.11$.**
- Phonon transitions are not vertical, so the relevant threshold is the *indirect* gap $\min E_\alpha - \max E_\beta$. That equals 2Δ only when $\bar{E}(s)$ is flat ($t_b = -t_a$); otherwise it is smaller. For photons, $\bar{E}$ cancels exactly.
- Away from the ring the quasiparticle velocity greatly exceeds c_s. Energy conservation then forces q almost tangent to the constant-energy contour.
- So phonons randomise momentum within an energy surface efficiently but relax energy slowly. Each event changes the energy by at most ω_D.
- The subsonic "passive" shell at the gap edge, the analogue of the polariton $E_0 = 2Mv_s^2$, has width $\sim 2.5 \times 10^{-4}\, t_a$ and is negligible.

Sources: phonon kernels.

- *Chang–Scalapino.* The superconducting phonon collision integral is written in energy, with an Ω^2 kernel, the factor

$1 - \Delta^2/EE'$ for scattering, $1 + \Delta^2/EE'$ for recombination, and recombination only for $\Omega \geq E + \Delta$ (Goldie–Withington, ar5iv 1208.0685).

- *Kaplan et al.* give the corresponding low-energy lifetimes (PRB 14, 4854 record).
- *Delta removal by a Jacobian.* Polariton and exciton papers remove the energy delta analytically: over q_z for 3D phonons coupled to a 2D layer (Tassone et al.; Hartwell thesis), or over $|q|$ for strictly 2D phonons, leaving the Jacobian $q_0/(\hbar c_{\mathrm{LA}})$ and a kinematic step function (Emary et al.).
- *Esin et al.* turn the q_z integral into a smooth weight $\propto \varepsilon\,\Theta(\varepsilon - \hbar v|q|)/\sqrt{\varepsilon^2 - \hbar^2 v^2 q^2}$ per (k, k') pair, so no delta function remains on the grid (arXiv:1710.09404).

Sources: Fokker–Planck validity.

- *Electrons in metals.* The degenerate Fokker–Planck form needs $\alpha^2 F$ to be "a narrow function on the temperature scale", i.e. $k_B T \gg \hbar\omega_D$ (Kabanov–Alexandrov, arXiv:0809.0818).
- *Hot-electron transport.* The energy-space FP form needs energy exchange to be "quasicontinuous" (arXiv:cond-mat/9909253).
- *2D quantum wells.* Explicit drift and diffusion coefficients for 2D acoustic phonons give $D/V \simeq T$ at 20 K (arXiv:cond-mat/0406360).
- *Equipartition regime.* Emission plus absorption collapses to $(2k_B T/\hbar\omega_q)\delta(\varepsilon - \varepsilon')$ only above the Bloch–Grüneisen temperature (Hwang–Das Sarma).
- *Near the gap edge.* Catelani and Basko keep the full phonon integral kernel and use a diffusion operator only for the photon term (arXiv:1807.07377).

Derivation: Fokker–Planck form and its validity. Kramers–Moyal expansion to second order of the Pauli equation, with detailed balance at T_c, gives

$$\nu_\lambda \partial_t n_\lambda = \partial_E\{\nu_\lambda D_\lambda[\partial_E n_\lambda + n_\lambda(1 - n_\lambda)/T_c]\},$$

with D_λ equal to half the second moment of the rate. Its zero-flux solutions are exactly $f((E - \mu_\lambda)/T_c)$ for any μ_λ. It requires three conditions:

1. $\omega_D \ll T_{\mathrm{cold}}$, i.e. a cold bath well above about 250 K for zone-wide transfers.
2. ω_D small compared with the scale on which n, ν and the coherence factors vary. This fails within about ω_D of the $E/\sqrt{E^2 - \Delta^2}$ edge.
3. **Sharp phonon lines.** With the damped-oscillator line, the emission tail $\sim 8\gamma\omega/\Omega^3$ gives a log-divergent second moment, so Lorentzian lines make FP formally invalid. They also open interband recombination at $\Omega = 2E_k$ with a rate $\propto \gamma\omega/(2E_k)^3$.

With typical line widths $\gamma_{\mathrm{ph}} = 0.1$–$0.3$ and $\omega_D \leq 0.22$, **broadening rather than c_s sets the energy-transfer scale**. As $c_s \to 0$ at finite γ, the bath tends to an Ohmic-like kernel that does not depend on c_s and is not elastic.

There is a controlled $c_s \to 0$ limit, but only for sharp lines with a $1/\omega_q$ coupling at fixed T_c:

- energy diffusion D stays finite and independent of c_s;
- the elastic, within-surface rate diverges as T_c/c_s^2;
- so energy-only FP becomes asymptotically exact. If the coupling lacks the $1/\omega_q$, instead, $D \propto c_s^2$ and the cold bath drops out of the NESS.

This bears directly on the planned "$c_s \to 0$" limit model: that limit is clean only with $\gamma_{\mathrm{ph}} \to 0$.

Derivation: when an energy-binned (s-binned) phonon equation is exact. It is exact if and only if, for every state k and every target shell J, the partial rates into and out of J depend on k only through s_k.

- *Exact:* momentum-independent vertices, the TM2 photon term, and all coherence factors, since Δ is k-independent and both bands are linear in s.
- *Violated:* the in-plane photon term (through $|X_k|^2$) and Peierls phonons (through $(\sin k - \sin k')^2$).

The research team computed the within-shell spread of $|X_k|^2$ on a 2048^2 grid:

s	**0**	**0.25**	**0.5**	**1.0**	**1.5**
$\langle\lvert X\rvert^2\rangle_s$	0.53	1.12	1.28	1.25	0.81
std/mean	1.25	0.48	0.31	0.14	0.06

The shell mean obeys the exact identity $\langle|X|^2\rangle_s = \int_s^2 s'\nu(s')ds'/\nu(s)$. The anisotropy is large near the van Hove shell. Its

effect on n is second order, because only rate ratios enter: $\delta n \approx (\partial n/\partial \ln r)\,\delta \ln r$, which vanishes when photon or phonon rates dominate. This matches the second-order shell-closure error and the $\leq 3 \times 10^{-5}$ shell-average shifts recorded in your `EI_bins_design.md`. On your Cartesian IBZ grid the point is moot, since $|X_k|^2$ is simply evaluated per k.

Inference: four phonon options, in order of preference.

1. **Exact energy-banded kernel.** Build the geometric pair density $P_{ij}(q)$ once by FFT cross-correlation of shell indicators. Then fill $W_{i\to j} \propto |M(q^*)|^2 P_{ij}(q^*)/c_s$ at $q^* = |E_i - E_j|/c_s$, and obtain reverse rates from detailed balance. This is essentially the FFT pair kernel already in `EI_bins`. The bin width must stay below $T_{\rm cold}/3$ and below typical transfers.
2. **Cell-averaged energy overlap on the Cartesian grid.** This is your current phonon route. It is acceptable because $\omega_q = c_s q$ varies by only about 0.005 across a cell. The electron energy spacing between neighbouring cells, though, is up to ~ 0.1–0.3 at $N = 64$, so the overlap or line width must exceed it.
3. **3D substrate phonons with q_z integrated (Esin type).** This gives smooth kernels on any grid, but it changes the physical model.
4. **Fokker–Planck, only where $\omega_D \ll T_{\rm cold}$.** Discretise it with a fermionic Chang–Cooper/Scharfetter–Gummel flux, $F_{i+1/2} = (gD/h)[B(x)n_i(1-n_{i+1}) - B(-x)n_{i+1}(1-n_i)]$ with $B(x) = x/(e^x - 1)$ and $x = h/T$. This flux vanishes exactly on Fermi–Dirac occupations on any grid, conserves number telescopically, and gives a tridiagonal rate matrix. It is the research team's construction, by analogy with the equilibrium-preserving Chang–Cooper scheme (Donnert–Brunetti).

No rank-one closed form exists, but four semi-analytic steady states do

Derivation: the general equations. With the vertical photon term and intraband-only phonons (sharp lines, $\omega_D < 2\Delta$):

$$\partial_t n_\alpha = -J_{\rm ph} + a_\alpha(1-n_\alpha) - b_\alpha n_\alpha, \qquad \partial_t n_\beta = +J_{\rm ph} + a_\beta(1-n_\beta) - b_\beta n_\beta,$$

where a_λ and b_λ are the phonon gain and loss functionals.

- The photon term conserves $n_\alpha + n_\beta$ at each k.
- The phonons conserve N_α and N_β.
- Together they conserve only the total N, so the combined NESS is unique at fixed filling.
- The vertical photon kernel is diagonal, the phonon kernel is banded in energy, and neither is rank one. **No global-(A, B) closed form exists.**

The structure still yields four semi-analytic steady states, described next.

First: an exact per-k quadratic, which generalises Eq. 16. Freeze a_λ, b_λ and write $R = \text{amp}\,\mathcal{R}/c^2, N = N_h(2E_k), n^0_\lambda = a_\lambda/(a_\lambda + b_\lambda)$ and $r_\lambda = R/(a_\lambda + b_\lambda)$. Stationarity at one k gives $n_\alpha = n^0_\alpha + r_\alpha J$ and $n_\beta = n^0_\beta - r_\beta J$, with

$$J = \frac{2J_0}{\mathsf{b} + \sqrt{\mathsf{b}^2 + 4r_\alpha r_\beta J_0}}, \quad J_0 = N n^0_\beta(1-n^0_\alpha) - (1+N)n^0_\alpha(1-n^0_\beta), \quad \mathsf{b} = 1 + r_\beta(N + n^0_\alpha) + r_\alpha(1 + N - n^0_\beta).$$

- Re-checked here against a numerical root of the 2×2 system for random parameters; it agrees to 10^{-9}. The $2J_0/(\mathsf{b} + \sqrt{\cdot})$ form avoids cancellation as $r \to 0$.
- **For $R = 0$ it reduces to $n = a/(a+b)$, which *is* Eq. 16 when a and b are separable.** The scheme is therefore the paper's iteration with A_j, B_j promoted to banded functions of energy.

Second: the fast-phonon quasi-Fermi limit, the analogue of Eq. 17. When phonon intraband relaxation outpaces photon pair exchange, $n_\lambda = f((E_\lambda - \mu_\lambda)/T_c)$ with two global constants. They are fixed by filling and by the global pair balance $\sum_k J_{\rm ph}(k) = 0$. In the non-degenerate case this is closed form:

$$e^{\Delta\mu/T_c} = \frac{\sum_k \mathcal{R}(k) N_h(2E_k)}{\sum_k \mathcal{R}(k)[1 + N_h(2E_k)]e^{-2E_k/T_c}}, \qquad \Delta\mu = \mu_\alpha - \mu_\beta.$$

- This gives $\Delta\mu = 0$ at $T_h = T_c$ and $\Delta\mu > 0$ for $T_h > T_c$, and it is **independent of every coupling strength**.
- For pumping concentrated at one energy E_0, it reduces to $\mu^* = E_0(1 - T_c/T_h)$ in the particle-hole-symmetric case, a Shockley–Queisser-type photon chemical potential.
- The unknowns are $(\Delta, m, \mu_\alpha, \mu_\beta)$, four scalars, the counterpart of (Δ, F).

If phonons *can* recombine pairs ($2\Delta_{\rm ind} < \omega_D$, or through Lorentzian tails), the Eliashberg-kinetics researcher showed that within this ansatz every phonon *scattering* bracket vanishes identically for any μ^*. The recombination bracket factorises as $(1-f)(1-f')n(\Omega)[e^{2\mu^*/T_c} - 1]$. The pair balance then gains a single cold term $L_0(\mu^*)[e^{2\mu^*/T_c} - 1]$ on the right side, which pulls $\Delta\mu$ toward zero.

Sources. This is the EI version of the Owen–Scalapino state, which treats pair-broken quasiparticles as "a Fermi gas at the

lattice temperature" in excess numbers because recombination is slower than thermalisation (PRL 28, 1559 record). Demsar notes that the Owen–Scalapino model predicts a first-order transition to the normal state with increasing excitation (arXiv:2003.11503). Elesin and Kopaev write the same state as $n = \{e^{(\epsilon-\mu^*)/T}+1\}^{-1}$. They warn that the quasi-equilibrium forms need recombination to be much slower than energy relaxation, a condition that degrades near the instability (Usp. Fiz. Nauk 133, 259). The Floquet two-bath steady states have the same structure: Fermi seas of electrons and holes with separate chemical potentials at the phonon temperature, and an excitation density scaling as the square root of the recombination rate (Seetharam et al.; Esin et al.).

Third and fourth: two explicit closures.

Photon-dominated limit. With particle–hole symmetry, the local balance $n^2/(1-n)^2 = e^{-2E/T_h}$ gives $n = f(E/T_h)$ exactly wherever $\mathcal{R} \neq 0$.

Relaxation-time closure. Replace the phonon term by $-\Gamma_c[n_\lambda - f((E_\lambda-\mu_\lambda)/T_c)]$, with μ_λ fixed by phonon band-number conservation. It is exact in both limits above and fully explicit in the particle–hole-symmetric case:

$$n(E) = \frac{-\mathsf{b}+\sqrt{\mathsf{b}^2+4\mathsf{ac}}}{2\mathsf{a}}, \quad \mathsf{a} = R,\ \mathsf{b} = \Gamma_c + 2RN_h,\ \mathsf{c} = \Gamma_c f_\mu(E) + RN_h.$$

It interpolates from f_μ at $R \to 0$ to $f(E/T_h)$ at $\Gamma_c \to 0$. It is the Curtis-style τ_{in} model (arXiv:1805.01482) without linearisation.

Two further reference solutions.

- In the $c_s \to 0$ equipartition limit, the full problem becomes a 1D nonlinear Fokker–Planck boundary-value problem in energy. It can be solved on about 10^3 points, like the paper's s-wave numerics.
- Replacing the acoustic phonons with a Lunkin-type flat cold bath gives an *exact* two-constant NESS through the per-k quadratic. That is an ideal regression test for the full solver.

Inference: the physics consequence.

No upward redistribution. Hot vertical photons only create and annihilate $\alpha\beta$ pairs; they cannot move quasiparticles in energy. Lunkin's flat $(+-)$ scattering is what shifts quasiparticles up "compared to the thermal distribution", and it has no counterpart here. Cold phonons do the opposite and pull quasiparticles down to the gap edge.

Gap bounds. In the fast-phonon limit with $T_h > T_c$, $\Delta\mu > 0$ implies $n \geq f(E/T_c)$, so $**\Delta_{\text{NESS}} < \Delta_{\text{eq}}(T_{\text{cold}})**$ for a monotone gap equation. If pumping sits at the gap edge ($E_0 = \Delta$), then $n = e^{-\Delta/T_h}e^{-(E-\Delta)/T_c} \leq e^{-E/T_h}$ for all $E \geq \Delta$. In that case the cold phonons can only help relative to the hot bath alone. When the step-like pumping extends above the edge, states between Δ and E_{eff} can exceed $f(E/T_h)$. Whether a finite gap survives when T_h exceeds the equilibrium T_c is a quantitative question, and the four-scalar quasi-Fermi model answers it cheaply.

Enhancement needs cold photons. Curtis et al. find gap enhancement for $\omega_0 > 2\Delta_0$ with a *cold* photon reservoir, through enhanced recombination (arXiv:1805.01482).

Bistability. Pumping concentrated near a threshold that moves with Δ, combined with the Owen–Scalapino tendency toward first-order collapse, could plausibly produce bistability. This is consistent with the subcritical coexistence your project already sees with sharp lines, but it remains speculative.

A prioritized plan for the existing grid code

The plan keeps your Cartesian grid, the C_{4v} IBZ, the s-window, both solvers and the existing phonon pathway, and confines the change to where the photon enters.

Priority	Change	Replaces	Acceptance test
1	Photon bath as a local vertical term: per-k $\Gamma^{\mathrm{abs}}, \Gamma^{\mathrm{em}}$ from `k0` / `k2`, added as $J_{\mathrm{ph}}(k)$	All photon entries of `_ltab` and the photon pair arrays	$T_h = T_c$ gives the thermal state to machine precision; $\Gamma^{\mathrm{abs}}/\Gamma^{\mathrm{em}} = e^{-2E/T_h}$; photon rates independent of N
2	Re-run the N-scan and old benchmark points	Nothing	NESS converges with N, limited by phonons only; quantify the shift of earlier phase maps
3	Phonons: keep the cell-averaged overlap on the grid, cross-check against the `EI_bins` FFT banded kernel inside the s-window; compute Δ_{ind} against ω_D; decide sharp versus Lorentzian lines	Nothing, initially	Number conservation; detailed balance; convergence in N and bin width; zero interband phonon rate for sharp lines when $2\Delta_{\mathrm{ind}} > \omega_D$
4	Solvers: per-k quadratic as the Picard local update; analytic 2×2 photon block in the Newton Jacobian; s-window set by a pumping threshold	The local μ target in Picard	Picard–Newton agreement; NESS insensitive to the window edges
5	Limit models: quasi-Fermi (four scalars), RTA closure, flat cold-bath exact benchmark, photon-only $f(E/T_h)$	Nothing	The full solver approaches each limit as the photon/phonon ratio is scanned

Priority 1 is a minimal change because the photon term becomes local. In the Picard μ-form, it adds to the existing gain and loss coefficients:

- $a_\alpha \mathrel{+}= \Gamma^{\mathrm{abs}} n_\beta$ and $b_\alpha \mathrel{+}= \Gamma^{\mathrm{em}}(1 - n_\beta)$;
- $a_\beta \mathrel{+}= \Gamma^{\mathrm{em}} n_\alpha$ and $b_\beta \mathrel{+}= \Gamma^{\mathrm{abs}}(1 - n_\alpha)$.

In Newton it adds one 2×2 block per k, with $\partial n/\partial\mu = -n(1-n)$.

Two pitfalls deserve explicit care:

- **Normalisation.** The local rate must enter as the *rate per state*. If it is inserted into a weighted pair sum $\sum_j w_j g_{ij}$, then $g_{\alpha_k\beta_k} = \Gamma^{\mathrm{abs}}(k)/w_k$. On a uniform grid that equals N_k times the $q = 0$ cell integral.
- **IBZ multiplicity.** In the IBZ, the star multiplicity carried by w_k must not multiply the vertical term, because each star member couples only to itself.

Adding $J_{\mathrm{ph}}(k)$ directly to the current, rather than through the pair table, avoids both. It also removes the dense photon pair arrays. That should ease the memory ceiling that pushed the full-BZ solver into the IBZ at $nk \approx 65$.

Priority 4 uses the exact local update:

```python
def loc_update(aa, ba, ab, bb, r, nh):
    # exact stationary (n_alpha, n_beta) at one k for frozen phonon gain/loss
    # aa, ba: phonon gain and loss coefficients of alpha; ab, bb: of beta
    # r = amp * R(k) / c**2 ; nh = N_h(2 E_k)
    na0, nb0 = aa / (aa + ba), ab / (ab + bb)
    ra, rb = r / (aa + ba), r / (ab + bb)
    j0 = nh * nb0 * (1 - na0) - (1 + nh) * na0 * (1 - nb0)
    bq = 1 + rb * (nh + na0) + ra * (1 + nh - nb0)
    j = 2 * j0 / (bq + np.sqrt(bq**2 + 4 * ra * rb * j0))
    return na0 + ra * j, nb0 - rb * j
```

What the code does:

- It returns the exact solution of the 2×2 local balance at fixed a_λ, b_λ.
- With `r = 0` it is your current target $n = a/(a+b)$.
- Convert the result back with $\mu = \ln[(1-n)/n]$. For deep tails, form $1 - n$ from the μ representation, as your μ-based Picard already does.

The s-window needs a new criterion, because the pumping is a step and not a ring. Every state with $2E_k > \omega_0$ is pumped at

a rate $\propto N_h(2E_k)\mathcal{R}(k)$. The window must therefore extend until $e^{-2E_k/T_h}\mathcal{R}/\mathcal{R}_{\max}$ falls below the solver tolerance. Freezing $n_\alpha = 0$ and $n_\beta = 1$ beyond that point is then consistent.

Finally, note what the slack variable covers. Your design notes record that with "no interband channel … two conserved populations; only one multiplier is implemented". In the photon-off limit, intraband-only phonons conserve N_α and N_β separately, so the photon-amplitude-to-zero continuation needs a second multiplier.

Assumptions to check against the progress notes

Dipole direction and mode thresholds. **An out-of-plane dipole at $z = L/2$ couples only through TM2, with threshold $2\omega_0$ and weight $1 - 4\omega_0^2/\Omega^2$ that vanishes at threshold.** At the antinode, E_z of TM1 vanishes (it goes as $\cos(\pi z/L)$), and so do the in-plane fields of TE2 and TM2. These polarization assignments are the research team's derivation, mapped onto Curtis's $\hat{\epsilon}_{1,2}$ vectors. Curtis et al. do not use TE/TM labels (arXiv:1805.01482v3).

Which channel is resonant. With $\omega_0 \approx 2\Delta$, the resonant $n = 1$ channel works only through the in-plane current vertex. Its interband element $(t_a - t_b)\Delta/E_k$ requires $t_a \neq t_b$. If the intent was resonant pumping *by the z-dipole*, ω_0 near Δ is the relevant design, and then the in-plane channel covers the whole band.

Gauge and Hartree terms in C_2. $C_2 = d(e_a - e_b)$ uses the bare H_0 in the current commutator. The velocity gauge with the mean-field H, and the length gauge with $\Delta\tau_x$ order and a $d\tau_x$ vertex, both give a rate $\propto \xi^2/E$ that vanishes at the ring. They agree with the bare form only if $e_a - e_b = 2\xi$, i.e. without Hartree or imbalance shifts. This matters little for TM2 above $2\omega_0$, where $\xi^2 \geq 3\Delta^2$, but it matters if ω_0 is lowered.

Normalisation. The normalisation chain must be confirmed against the progress-note equations:

- the $1/N_k$ per pair;
- amp $= Ne^2/(2V\epsilon_0)$ with the $2/L$ mode factor;
- the 2π convention, in which "the line carries the full factor";
- any mode-profile factor for TM2 beyond $(cq/\omega_{2q})^2$.

Kernel assumptions. The kernels assume:

- a q-independent γ (a $\gamma(\omega)$ turns K_0, K_2 into a cheap 1D quadrature);
- equal amplitudes for TE1, TM1 and TM2;
- modes extending to $q \to \infty$, which is harmless because the integrals converge.

GGE validity. The GGE requires all system–bath rates, not the cavity loss γ, to stay well below Δ. This weakens as $\Delta \to 0$.

Provenance.

From your project, not checked against code. The code details come from the task description, the Magistrska project documents (`EI_grid_design.md` , `EI_bins_design.md`) and stored project notes: the IBZ move, the s-window, the `EI_limit` limit-model plan, the z-dipole choice, the planned Peierls-diagonal phonon vertex, the NESS gap values and the observed bistability. None of these was checked against the source code, and no researcher read `progress_notes.pdf` . The vertex forms above therefore rest on the design documents.

Citations that are unverified or partly reconstructed:

- Owen–Scalapino authorship and content rest on the abstract only.
- Parker's PRB 12, 3667 and the author lists for the Tikhonov et al. and Catelani–Basko papers were recalled from memory. The Semenov et al. attribution likewise came from memory.
- The Elesin–Kopaev excitonic-insulator sign parameter comes from an OCR-garbled scan.
- The Tassone Eq. 3 prefactor was reconstructed from OCR.
- The Chang–Cooper original was cited second-hand.
- The Ulčakar–Lenarčič GGE paper could not be read.
- Several arXiv journal references (2509.07865, 2512.13065, 2401.12607, cond-mat/0207547) were not confirmed.
- One research thread read Lunkin et al. only from the local text copy.

Derivations with no published counterpart. No published source treats a clean excitonic insulator with a gapped planar-cavity bath and an acoustic-phonon bath in a kinetic or GGE framework. These are unpublished derivations:

- the vertical reduction;
- the K_0/K_2 closed forms;
- the per-k quadratic;
- the quasi-Fermi pair-balance formula;
- the fermionic Chang–Cooper flux.

Conclusion

The narrow photon line is not a resolution problem; it is a modelling problem. The physical quantity is the photon factor integrated over the $q = 0$ cell. That integral is exactly analytic, and it shows that a planar cavity is a step absorber, not a resonance. Brute-force convergence would need about 5×10^5 points per axis. Meanwhile, the grid-sampled version has made the photon/phonon ratio, the only coupling quantity the NESS remembers, an artefact of N. Earlier Cartesian phase maps should be read as runs at an unknown, N-dependent effective coupling ratio with a ring-shaped rather than step-shaped pumping profile. This is the most consequential correction the analysis offers.

The analytic reduction also changes the physical picture compared with Lunkin et al. Their stabilisation depends on a hot bath that pushes quasiparticles up in energy. Vertical hot photons cannot do that, and quasi-elastic cold phonons that cannot recombine pairs (since $2\Delta > \omega_D$) hand control of the pair number to the photon temperature. The result is an Owen–Scalapino μ^* state whose gap is bounded by the cold-bath equilibrium value and which is prone to first-order behaviour. Two routes therefore look most promising for non-equilibrium enhancement in this geometry. One is to invert the temperatures so the photon bath is colder than the phonons, as Curtis et al. found. The other is to add an intraband redistribution channel that vertical photons lack. Both can be screened with the four-scalar quasi-Fermi model before any full-grid run.